Student Performance Analysis using PostgreSQL

Project Overview

This project analyzes student performance data using PostgreSQL. The objective is to identify factors that influence academic performance and gain insights from student behavior, study habits, and demographic information.

Dataset Information

Dataset: Student Performance Dataset

Total Records: 649 Students

The dataset contains information related to:

- Student demographics
- Study habits
- Family support
- Internet access
- Alcohol consumption
- Academic performance

Project Objectives

The project aims to answer the following business questions:

1. How many students are in the dataset?
2. Which gender performs better academically?
3. Does study time improve exam performance?
4. How do previous failures affect grades?
5. Does internet access impact student performance?
6. Does family support improve academic results?
7. How can students be ranked based on exam scores?

SQL Skills Used

- Data Cleaning
- ALTER TABLE
- Column Renaming
- Aggregate Functions (COUNT, AVG)
- GROUP BY
- ORDER BY
- LIMIT & OFFSET
- Subqueries
- Window Functions (ROW_NUMBER)

Key Insights

- Students with higher study time generally achieved better final exam scores.
- Previous academic failures negatively impacted performance.
- Internet access showed a positive relationship with academic results.
- Family support contributed to improved student performance.
- Window functions helped rank students according to exam scores.

Conclusion

This project demonstrates practical SQL skills used in real-world data analysis. Through data cleaning, aggregation, subqueries, and window functions, valuable insights were extracted from student performance data. The project strengthened my understanding of SQL and business-oriented data analysis.

GitHub Repository:
https://github.com/takale-Vaibhavi/Student-Performance-SQL-Project